OPERATING SYSTEMS PRACTICAL

Experiment: FCFS Disk Scheduling

1. Aim

To write and execute a C program to implement First Come First Serve (FCFS) disk scheduling and calculate the total head movement.

2. Theory

First Come First Serve (FCFS) is a disk scheduling algorithm in which disk I/O requests are serviced in the same order in which they arrive in the request queue. The disk head starts from the given initial position and moves to each request one by one, without changing their order.

The head movement for each request is calculated by finding the absolute difference between the current head position and the requested cylinder. The total head movement is the sum of all these individual movements. FCFS is simple and fair because requests are handled in arrival order, but it may result in large total head movement and higher seek time.

3. Algorithm

1. Start the program.
2. Read the number of disk requests.
3. Read all cylinder numbers in the disk request queue.
4. Read the initial disk head position.
5. Set the current head position equal to the initial head position and total head movement to zero.
6. Display the initial head position as the beginning of the seek sequence.
7. For each request in the given order, calculate movement = |request[i] – current|.
8. Add the movement to total head movement and update current = request[i].
9. Display each visited cylinder in the seek sequence.
10. Display the total head movement in cylinders.
11. Stop the program.

4. C Program

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, head, i;
    int totalHeadMovement = 0;
    int current;

    printf("Enter the number of disk requests: ");
    scanf("%d", &n);

    int request[n];

    printf("Enter the disk request queue: ");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &request[i]);
    }

    printf("Enter the initial head position: ");
    scanf("%d", &head);

    current = head;

    printf("\\nFCFS Disk Scheduling\\n");
    printf("Seek Sequence: %d", head);

    for(i = 0; i < n; i++)
    {
        int movement = abs(request[i] - current);

        totalHeadMovement = totalHeadMovement + movement;

        current = request[i];

        printf(" -> %d", current);
    }

    printf("\\nTotal Head Movement = %d cylinders\\n",
           totalHeadMovement);

    return 0;
}
```

5. Formula / Calculation

Term	Formula / Meaning
Head Movement	Movement = |Next cylinder – Current head position|
Total Head Movement	Sum of the movement for all requests

Absolute difference is used because head movement is counted as a positive number of cylinders, regardless of the direction of movement.

6. Sample Input and Output

Sample input:

```
Enter the number of disk requests: 5
Enter the disk request queue: 90 30 100 60 80
Enter the initial head position: 50
```

Calculation:

Move	Calculation	Head movement
50 → 90	|90 – 50|	40
90 → 30	|30 – 90|	60
30 → 100	|100 – 30|	70
100 → 60	|60 – 100|	40
60 → 80	|80 – 60|	20
	Total	**230 cylinders**

Expected output:

```
FCFS Disk Scheduling
Seek Sequence: 50 -> 90 -> 30 -> 100 -> 60 -> 80
Total Head Movement = 230 cylinders
```

7. Conclusion

Thus, the C program for FCFS disk scheduling was implemented and executed successfully. The disk requests were serviced in the given order, the seek sequence was displayed, and the total head movement was calculated as 230 cylinders for the sample input.

Important Note

FCFS services requests strictly in the order in which they appear in the request queue. It does not rearrange requests to reduce head movement. Therefore, the total head movement depends on the order of the requests and the initial head position.